

Calculus

• Derivative of Trigonometric Functions.

$$> \frac{d(\sin x)}{dx} = \cos x$$

$$> \frac{d(\operatorname{cosec} x)}{dx} = -\operatorname{cosec} x \cdot \cot x$$

$$> \frac{d(\cos x)}{dx} = -\sin x$$

$$> \frac{d(\sec x)}{dx} = \sec x \cdot \tan x$$

$$> \frac{d(\tan x)}{dx} = \sec^2 x$$

$$> \frac{d(\cot x)}{dx} = -\operatorname{cosec}^2 x$$

• Derivative of Exponential and logarithmic Function:

$$> \frac{d(e^x)}{dx} = e^x$$

$$> \frac{d(a^x)}{dx} = a^x (\ln a); a > 0 \text{ but } a \neq 1$$

$$> \frac{d(\ln x)}{dx} = \frac{1}{x}$$

$$> \frac{d(\log_a x)}{dx} = \frac{1}{x} \log_a e; a > 0 \text{ but } a \neq 1$$

• Derivative of Inverse Trigonometric Functions.

$$> \frac{d(\sin^{-1} x)}{dx} = \frac{1}{\sqrt{1-x^2}} \quad (\text{for } |x| < 1)$$

$$> \frac{d(\tan^{-1} x)}{dx} = \frac{1}{x^2+1}$$

$$> \frac{d(\sec^{-1} x)}{dx} = \frac{1}{x\sqrt{x^2-1}}$$

$$> \frac{d(\cos^{-1} x)}{dx} = \frac{-1}{\sqrt{1-x^2}} \quad (\text{for } |x| < 1)$$

$$> \frac{d(\cot^{-1}x)}{dx} = \frac{-1}{1+x^2}$$

$$> \frac{d(\operatorname{cosec}^{-1}x)}{dx} = \frac{-1}{x\sqrt{x^2-1}} \quad (\text{for } |x| > 1)$$

• Hyperbolic Function

→ The hyperbolic Functions are defined as follows:

For any real number x , $\sinh x = \frac{e^x - e^{-x}}{2}$

$$> \cosh x = \frac{e^x + e^{-x}}{2} \quad \text{Where,}$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots$$

Also,

$$> \tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$> \operatorname{cosech} x = \frac{2}{e^x - e^{-x}}$$

$$> \operatorname{sech} x = \frac{2}{e^x + e^{-x}}$$

$$> \coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

→ Some results

$$\textcircled{1} \sinh 2x = 2 \sinh x \cdot \cosh x.$$

$$\textcircled{2} \cosh^2 x + \sinh^2 x = \cosh 2x$$

$$\textcircled{3} \cosh^2 x - \sinh^2 x = 1$$

$$\textcircled{4} \operatorname{sech}^2 x + \tanh^2 x = 1$$

⑤ $\coth^2 x - \operatorname{cosech}^2 x = 1$

• Derivative of Hyperbolic Function.

① Derivative of $\sinh x$.

let $y = \sinh x$.

differentiating both sides with respect to x .

$$\frac{dy}{dx} = \frac{d \sinh x}{dx}$$

$$= \frac{d \left(\frac{e^x - e^{-x}}{2} \right)}{dx}$$

$$= \frac{1}{2} \frac{d(e^x - e^{-x})}{dx}$$

$$= \frac{1}{2} \left(\frac{de^x}{dx} - \frac{de^{-x}}{dx} \right)$$

$$= \frac{1}{2} (e^x + e^{-x})$$

$$= \frac{e^x + e^{-x}}{2}$$

$$\therefore \frac{d \sinh x}{dx} = \cosh x$$

$$\textcircled{2} \therefore \frac{d \cosh x}{dx} = \sinh x$$

$\Rightarrow$ let $y = \cosh x$

② differentiating both sides with respect to x .

$$\frac{dy}{dx} = \frac{d \cosh x}{dx}$$

$$= \frac{d \left(\frac{e^x + e^{-x}}{2} \right)}{dx}$$

$$= \frac{1}{2} \left(\frac{de^n}{dn} + \frac{de^{-n}}{dn} \right)$$

$$= \frac{1}{2} (e^n - e^{-n})$$

$$= \frac{e^n - e^{-n}}{2}$$

$$= \sinh n$$

③ Derivative of $\tanh n$.

let $y = \tanh n$

differentiating both sides with respect to n .

$$\frac{dy}{dn} = \frac{d \tanh n}{dn}$$

$$= \frac{d \left(\frac{e^n - e^{-n}}{e^n + e^{-n}} \right)}{dn}$$

$$= \frac{(e^n + e^{-n}) \frac{d(e^n - e^{-n})}{dn} - (e^n - e^{-n}) \frac{d(e^n + e^{-n})}{dn}}{(e^n + e^{-n})^2}$$

$$= \frac{(e^n + e^{-n})(e^n + e^{-n}) - (e^n - e^{-n})(e^n - e^{-n})}{(e^n + e^{-n})^2}$$

$$= \frac{(e^n + e^{-n})^2 - (e^n - e^{-n})^2}{(e^n + e^{-n})^2}$$

$$= \frac{(e^n + e^{-n} + e^n - e^{-n})(e^n + e^{-n} - e^n + e^{-n})}{(e^n + e^{-n})^2}$$

$$= \frac{2e^n \times 2e^{-n}}{(e^n + e^{-n})^2}$$

$$= \frac{4}{(e^n + e^{-n})^2}$$

$$= \left(\frac{2}{e^n + e^{-n}} \right)^2$$

$$= \operatorname{sech}^2 n$$

④ Derivative of cosech x

let $y = \text{cosech } x$

differentiating both sides with respect to x .

$$\frac{dy}{dx} = \frac{d(\text{cosech } x)}{dx}$$

$$= \frac{d\left(\frac{2}{e^x - e^{-x}}\right)}{dx}$$

$$= \frac{(e^x - e^{-x}) \frac{d2}{dx} - 2 \frac{d(e^x - e^{-x})}{dx}}{(e^x - e^{-x})^2}$$

$$= \frac{(e^x - e^{-x}) \times 0 - 2(e^x + e^{-x})}{(e^x - e^{-x})^2}$$

$$= \frac{-2(e^x + e^{-x})}{(e^x - e^{-x})^2}$$

$$= \frac{-2}{(e^x - e^{-x})} \times \left(\frac{e^x + e^{-x}}{e^x - e^{-x}}\right)$$

$$= -\text{cosech } x \times \text{coth } x$$

⑤ Derivative of sech x

let $y = \text{sech } x$.

differentiating both sides with respect to x .

$$\frac{dy}{dx} = \frac{d(\text{sech } x)}{dx}$$

$$= \frac{d\left(\frac{2}{e^x + e^{-x}}\right)}{dx}$$

$$= \frac{(e^x + e^{-x}) \frac{d2}{dx} - 2 \frac{d(e^x + e^{-x})}{dx}}{(e^x + e^{-x})^2} \times 1$$

$$= \frac{(e^x + e^{-x}) \times 0 - 2(e^x - e^{-x})}{(e^x + e^{-x})^2}$$

$$= \frac{-2(e^n - e^{-n})}{(e^n + e^{-n})^2}$$

$$= \frac{-2}{(e^n + e^{-n})} \times \frac{(e^n - e^{-n})}{e^n + e^{-n}}$$

$$= -\operatorname{sech} n \cdot \tanh n$$

⑥ Derivative of $\coth n$.

let $y = \coth n$.

differentiating both sides with respect to n .

$$\frac{dy}{dn} = \frac{d \coth n}{dn}$$

$$= \frac{d \left(\frac{e^n + e^{-n}}{e^n - e^{-n}} \right)}{dn}$$

$$= \frac{(e^n - e^{-n}) \frac{d(e^n + e^{-n})}{dn} - (e^n + e^{-n}) \times \frac{d(e^n - e^{-n})}{dn}}{(e^n - e^{-n})^2}$$

$$= \frac{(e^n - e^{-n})(e^n - e^{-n}) - (e^n + e^{-n})(e^n + e^{-n})}{(e^n - e^{-n})^2}$$

$$= \frac{(e^n - e^{-n})^2 - (e^n + e^{-n})^2}{(e^n - e^{-n})^2}$$

$$= \frac{(e^n - \cancel{e^n} + e^n + \cancel{e^n})(\cancel{e^n} - e^{-n} - \cancel{e^n} - e^{-n})}{(e^n - e^{-n})^2}$$

$$= \frac{2e^n \times (-2e^{-n})}{(e^n - e^{-n})^2}$$

$$= - \left(\frac{2}{e^n - e^{-n}} \right)^2$$

$$= -\operatorname{cosech}^2 n$$

• Derivative of ~~sin~~ Inverse Hyperbolic Function

① Derivative of $\sinh^{-1} x$.

$$\text{let } y = \sinh^{-1} x.$$

$$\Rightarrow \sinh y = x$$

$$\Rightarrow x = \sinh y.$$

diff. both sides with respect to y .

$$\frac{dx}{dy} = \frac{d \sinh y}{dy}$$

$$\Rightarrow \frac{dx}{dy} = \cosh y$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\cosh y}.$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{1 + \sinh^2 y}} \quad [\because \cosh^2 x - \sinh^2 x = 1]$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{1 + x^2}}$$

$$\therefore \frac{d(\sinh^{-1} x)}{dx} = \frac{1}{\sqrt{x^2 + 1}}$$

② Derivative of $\cosh^{-1} x$.

$$\text{let } y = \cosh^{-1} x.$$

$$\Rightarrow \cosh y = x$$

$$\Rightarrow x = \cosh y.$$

diff. both sides with respect to y .

$$\frac{dx}{dy} = \frac{d \cosh y}{dy}$$

$$\Rightarrow \frac{dx}{dy} = \sinh y$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sinh y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{\cosh^2 x - 1}} \quad [\because \cosh^2 x - \sinh^2 x = 1]$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{x^2 - 1}}$$

$$\therefore \frac{d(\cosh^{-1} x)}{dx} = \frac{1}{\sqrt{x^2 - 1}}$$

③ Derivative of $\operatorname{sech}^{-1} x$.

$$\text{let } y = \operatorname{sech}^{-1} x.$$

$$\Rightarrow \operatorname{sech} y = x$$

$$\Rightarrow x = \operatorname{sech} y$$

diff. both sides with respect to y .

$$\frac{dx}{dy} = \frac{d(\operatorname{sech} y)}{dy}$$

$$\frac{dx}{dy} = -\operatorname{sech} y \cdot \tanh y$$

$$\Rightarrow \frac{dy}{dx} = \frac{-1}{\operatorname{sech} y \cdot \tanh y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-1}{x \sqrt{1 - \operatorname{sech}^2 x}} \quad [\operatorname{sech}^2 x + \tanh^2 x = 1]$$

$$\Rightarrow \frac{dy}{dx} = \frac{-1}{x \sqrt{1 - x^2}}$$

$$\Rightarrow \frac{d(\operatorname{sech}^{-1} x)}{dx} = \frac{-1}{x \sqrt{1 - x^2}}$$

④ Derivative of $\tanh^{-1} x$.

$$\text{let } y = \tanh^{-1} x.$$

$$\Rightarrow \tanh y = x.$$

$$\Rightarrow x = \tanh y.$$

diff. both sides with respect to y .

$$\frac{dx}{dy} = \frac{d(\tanh y)}{dy}$$

$$\Rightarrow \frac{dn}{dy} = \operatorname{sech}^2 y$$

$$\Rightarrow \frac{dy}{dn} = \frac{1}{\operatorname{sech}^2 y}$$

$$\Rightarrow \frac{dy}{dn} = \frac{1}{1 - \tanh^2 n} \quad [\operatorname{sech}^2 n + \tanh^2 n = 1]$$

$$\Rightarrow \frac{dy}{dn} = \frac{1}{1 - n^2}$$

$$\therefore \frac{d \tanh^{-1} n}{dn} = \frac{1}{1 - n^2}$$

⑤ derivative of $\operatorname{cosech}^{-1} n$.

let $y = \operatorname{cosech}^{-1} n$.

$$\Rightarrow \operatorname{cosech} y = n$$

$$\Rightarrow n = \operatorname{cosech} y$$

diff. both sides with respect to y .

$$\frac{dn}{dy} = \frac{d(\operatorname{cosech} y)}{dy}$$

$$\Rightarrow \frac{dn}{dy} = -\operatorname{cosech} y \cdot \coth y$$

$$\Rightarrow \frac{dy}{dn} = \frac{-1}{\operatorname{cosech} y \cdot \coth y}$$

$$= \frac{-1}{n \sqrt{1 + \operatorname{cosech}^2 y}} \quad [\because \coth^2 n - \operatorname{cosech}^2 y = 1]$$

$$= \frac{-1}{n \sqrt{1 + n^2}}$$

$$\therefore \frac{d(\operatorname{cosech}^{-1} n)}{dn} = \frac{-1}{n \sqrt{1 + n^2}}$$

⑥ Derivative of $\coth^{-1} x$.

let $y = \coth^{-1} x$.

$\Rightarrow \coth y = x$

$\Rightarrow x = \coth y$.

diff. both sides with respect to y .

$$\frac{dx}{dy} = \frac{d \coth y}{dy}$$

$$\Rightarrow \frac{dx}{dy} = -\operatorname{cosech}^2 y$$

$$\Rightarrow \frac{dy}{dx} = \frac{-1}{\operatorname{cosech}^2 y}$$

$$= \frac{-1}{\coth^2 x - 1} \quad [\because \coth^2 x - \operatorname{cosech}^2 x = 1]$$

$$= \frac{-1}{x^2 - 1}$$

$$\therefore \frac{d(\coth^{-1} x)}{dx} = \frac{-1}{x^2 - 1}$$

• Exercise 13.1

Part B: Creative Questions

① Find the derivatives with respect to x of the followings:

a) $\cosh 5x$.

let $y = \cosh 5x$.

differentiating both sides with respect to x .

$$\frac{dy}{dx} = \frac{d \cosh 5x}{dx}$$

$$= \frac{d \cosh 5x}{d 5x} \times \frac{d 5x}{dx}$$

$$= \sinh 5x \times 5$$

$$= 5 \sinh 5x$$

b) $\sinh \frac{x^2}{a}$

let $y = \sinh \frac{x^2}{a}$

differentiating both sides w.r.t x .

$$\frac{dy}{dx} = \frac{d \left(\sinh \frac{x^2}{a} \right)}{dx}$$

$$= \frac{d \sinh \frac{x^2}{a}}{d \frac{x^2}{a}} \times \frac{d \frac{x^2}{a}}{dx}$$

$$= \cosh \frac{x^2}{a} \times \frac{1}{a} \times 2x$$

$$= \frac{2x}{a} \cosh \frac{x^2}{a}$$

c) $\sinh^2 \frac{n}{a}$

let $y = \sinh^2 \frac{n}{a}$

differentiating both sides w.r.t n .

$$\frac{dy}{dn} = \frac{d\left(\sinh^2 \frac{n}{a}\right)}{dn}$$

$$= \frac{d\left(\sinh^2 \frac{n}{a}\right)}{d\left(\sinh \frac{n}{a}\right)} \times \frac{d\left(\sinh \frac{n}{a}\right)}{d\left(\frac{n}{a}\right)} \times \frac{d\left(\frac{n}{a}\right)}{dn}$$

$$= 2 \sinh \frac{n}{a} \times \cosh \frac{n}{a} \times \frac{1}{a}$$

$$= \frac{1}{a} \sinh \frac{2n}{a} \quad [\because \sinh 2n = 2 \sinh n \cdot \cosh n]$$

d) $\tanh^2 \frac{n}{2a}$

let $y = \tanh^2 \frac{n}{2a}$

differentiating both sides w.r.t n .

$$\frac{dy}{dn} = \frac{d\left(\tanh^2 \frac{n}{2a}\right)}{dn}$$

$$= \frac{d\left(\tanh^2 \frac{n}{2a}\right)}{d\left(\tanh \frac{n}{2a}\right)} \times \frac{d\left(\tanh \frac{n}{2a}\right)}{d\left(\frac{n}{2a}\right)} \times \frac{d\left(\frac{n}{2a}\right)}{dn}$$

$$= 2 \tanh \frac{n}{2a} \times \operatorname{sech}^2 \frac{n}{2a} \times \frac{1}{2a}$$

$$= \frac{1}{a} \tanh \frac{n}{2a} \times \operatorname{sech}^2 \frac{n}{2a}$$

e) $\cosh^{-1} 2x$

let $y = \cosh^{-1} 2x$

differentiating both sides w.r.t x .

$$\frac{dy}{dx} = \frac{d \cosh^{-1} 2x}{dx}$$

$$= \frac{d \cosh^{-1} 2x}{d 2x} \times \frac{d 2x}{dx}$$

$$= \frac{1}{\sqrt{(2x)^2 - 1}} \times 2$$

$$= \frac{2}{\sqrt{4x^2 - 1}}$$

f) $\sinh^{-1} ax$

let $y = \sinh^{-1} ax$

differentiating both sides w.r.t x .

$$\frac{dy}{dx} = \frac{d \sinh^{-1} ax}{dx}$$

$$= \frac{d \sinh^{-1} ax}{d ax} \times \frac{d ax}{dx}$$

$$= \frac{1}{\sqrt{a^2 x^2 + 1}} \times a$$

$$= \frac{a}{\sqrt{a^2 x^2 + 1}}$$

g) $\operatorname{sech}^{-1} 3x$

let $y = \operatorname{sech}^{-1} 3x$

differentiating both sides w.r.t x .

$$\frac{dy}{dx} = \frac{d \operatorname{sech}^{-1} 3x}{dx}$$

$$= \frac{d \operatorname{sech}^{-1} 3x}{d 3x} \times \frac{d 3x}{dx}$$

$$= \frac{-1}{3x \sqrt{1 - (3x)^2}} \times 3$$

$$= \frac{-1}{x\sqrt{1-9x^2}}$$

h) $\operatorname{cosech}^{-1} 2x$

let $y = \operatorname{cosech}^{-1} 2x$

differentiating both sides w.r.t x .

$$\frac{dy}{dx} = \frac{d(\operatorname{cosech}^{-1} 2x)}{dx}$$

$$= \frac{d(\operatorname{cosech}^{-1} 2x)}{d2x} \times \frac{d2x}{dx}$$

$$= \frac{-1}{2x\sqrt{1+(2x)^2}} \times 2$$

$$= \frac{-1}{x\sqrt{1+4x^2}}$$

i) $\tanh^{-1} x/2$

let $y = \tanh^{-1} x/2$

differentiating both sides w.r.t x .

$$\frac{dy}{dx} = \frac{d(\tanh^{-1} x/2)}{dx}$$

$$= \frac{d(\tanh^{-1} x/2)}{d(x/2)} \times \frac{d(x/2)}{dx}$$

$$= \frac{1}{1-(\frac{x}{2})^2} \times \frac{1}{2}$$

$$= \frac{1}{1-\frac{x^2}{4}} \times \frac{1}{2}$$

$$= \frac{4 \times 2}{(4-x^2)} \times \frac{1}{2}$$

$$= \frac{2}{4-x^2}$$

∴

② Find the differential coefficients of the following with respect to x .

a) $\operatorname{sech}^{-1} x - \operatorname{cosech}^{-1} x = y$ (let)

$$\Rightarrow \frac{d(\operatorname{sech}^{-1} x - \operatorname{cosech}^{-1} x)}{dx} = \frac{dy}{dx}$$

$$= \frac{d(\operatorname{sech}^{-1} x)}{dx} - \frac{d(\operatorname{cosech}^{-1} x)}{dx}$$

$$= \frac{-1}{x\sqrt{1-x^2}} - \left(\frac{-1}{x\sqrt{1+x^2}} \right)$$

$$= \frac{-1}{x\sqrt{1-x^2}} + \frac{1}{x\sqrt{1+x^2}}$$

$$= \frac{1}{x} \left[\frac{1}{\sqrt{1+x^2}} - \frac{1}{\sqrt{1-x^2}} \right]$$

b) $\sinh^2 \frac{x}{a} + \cosh^2 \frac{x}{a} = y$ (let)

$$\Rightarrow \frac{d(\sinh^2 \frac{x}{a})}{dx} + \frac{d(\cosh^2 \frac{x}{a})}{dx} = \frac{dy}{dx}$$

$$= \frac{d(\sinh^2 \frac{x}{a})}{d(\sinh \frac{x}{a})} \times \frac{d(\sinh \frac{x}{a})}{d(\frac{x}{a})} \times \frac{d(\frac{x}{a})}{dx} +$$

$$\frac{d(\cosh^2 \frac{x}{a})}{d(\cosh \frac{x}{a})} \times \frac{d(\cosh \frac{x}{a})}{d(\frac{x}{a})} \times \frac{d(\frac{x}{a})}{dx}$$

$$= 2 \sinh \frac{x}{a} \times \cosh \frac{x}{a} \times \frac{1}{a} + 2 \cosh \frac{x}{a} \times \sinh \frac{x}{a} \times \frac{1}{a}$$

$$= \frac{2}{a} \left[\sinh \frac{x}{a} \times \cosh \frac{x}{a} + \sinh \frac{x}{a} \times \cosh \frac{x}{a} \right]$$

$$= \frac{2}{a} \left[2 \sinh \frac{x}{a} \cdot \cosh \frac{x}{a} \right]$$

$$= \frac{2}{a} \sinh \frac{2x}{a}$$

$$c) \frac{L}{\sinh x + \cosh x} = y \text{ (let)}$$

$$\Rightarrow \frac{d}{dx} \left(\frac{L}{\sinh x + \cosh x} \right) = \frac{dy}{dx}$$

$$= (\sinh x + \cosh x) \frac{dL}{dx} - L \frac{d(\sinh x + \cosh x)}{dx}$$

$$= \frac{(\sinh x + \cosh x)^2}{(\sinh x + \cosh x)^2} \times 0 - (\cosh x + \sinh x)$$

$$= - \frac{(\cosh x + \sinh x)}{(\sinh x + \cosh x)^2}$$

$$= - \frac{L}{(\cosh x + \sinh x)} \times \frac{(\cosh x - \sinh x)}{(\cosh x - \sinh x)}$$

$$= - \frac{(\cosh x - \sinh x)}{\cosh^2 x - \sinh^2 x}$$

$$= \frac{\sinh x - \cosh x}{L} \quad [\because \cosh^2 x - \sinh^2 x = 1]$$

$$= \sinh x - \cosh x$$

$$d) \sqrt{\sinh x + \cosh x} = y \text{ (let)}$$

$$\Rightarrow \frac{d(\sinh x + \cosh x)^{1/2}}{dx} = \frac{dy}{dx}$$

$$= \frac{d(\sinh x + \cosh x)^{1/2}}{d(\sinh x + \cosh x)} \times \frac{d(\sinh x + \cosh x)}{dx}$$

$$= \frac{1}{2} (\sinh x + \cosh x)^{-1/2} \times (\cosh x + \sinh x)$$

$$= \frac{1}{2} \times \frac{1}{(\sinh x + \cosh x)^{1/2}} \times (\cosh x + \sinh x)$$

$$= \frac{1}{2} \times \frac{(\sqrt{\sinh x + \cosh x})^2}{\sqrt{\sinh x + \cosh x}}$$

$$= \frac{1}{2} (\sinh x + \cosh x)^{1/2}$$

3. Find the derivatives (with respect to x) of the followings:

a) $\ln(\tanh x) \Rightarrow y = \ln(\tanh x)$

$\Rightarrow$ differentiating both sides with respect to x .

$$\begin{aligned}\frac{dy}{dx} &= \frac{d(\ln(\tanh x))}{dx} \\ &= \frac{d(\ln(\tanh x))}{d(\tanh x)} \times \frac{d(\tanh x)}{dx} \\ &= \frac{1}{\tanh x} \times \sec^2 x \\ &= \frac{\cosh x}{\sinh x} \times \frac{1}{\cosh x \times \cosh x} \\ &= \frac{2}{2 \sinh x \cdot \cosh x} \\ &= \frac{2}{\sinh 2x} \\ &= 2 \operatorname{cosech} 2x\end{aligned}$$

b) $\tanh(\arcsin x)$

$\Rightarrow$ let $y = \tanh(\arcsin x)$

differentiating both sides with respect to x .

$$\begin{aligned}\frac{dy}{dx} &= \frac{d(\tanh(\arcsin x))}{dx} \\ &= \frac{d(\tanh(\arcsin x))}{d(\arcsin x)} \times \frac{d(\arcsin x)}{dx} \\ &= \sec^2(\arcsin x) \times \frac{d(\sin^{-1} x)}{dx} \\ &= \sec^2(\sin^{-1} x) \times \frac{1}{\sqrt{1-x^2}}\end{aligned}$$

c) $2 \operatorname{arc} \tan \left(\tanh \frac{\eta}{2} \right)$

Let $y = 2 \operatorname{arc} \tan \left(\tanh \frac{\eta}{2} \right)$

differentiating on both sides with respect to η .

$$\frac{dy}{d\eta} = \frac{d}{d\eta} 2 \operatorname{arc} \tan \left(\tanh \frac{\eta}{2} \right)$$

$$= 2 \frac{d \tan^{-1} \left(\tanh \frac{\eta}{2} \right)}{d \tanh \frac{\eta}{2}} \times \frac{d \tanh \frac{\eta}{2}}{d \frac{\eta}{2}} \times \frac{d \frac{\eta}{2}}{d \eta}$$

$$= \cancel{2} \times \frac{1}{1 + \tanh^2 \frac{\eta}{2}} \times \operatorname{sech}^2 \frac{\eta}{2} \times \frac{1}{\cancel{2}}$$

$$= \frac{1}{\cosh^2 \frac{\eta}{2} + \sinh^2 \frac{\eta}{2}}$$

$$= \frac{1}{\cosh^2 \frac{\eta}{2}} \times \frac{\cosh^2 \frac{\eta}{2}}{\sinh^2 \frac{\eta}{2} + \cosh^2 \frac{\eta}{2}}$$

$$= \frac{1}{\cosh 2 \times \frac{\eta}{2}}$$

$$= \frac{1}{\cosh \eta}$$

$$= \operatorname{sech} \eta$$

d) $\cosh^{-1}(\sinh 2x)$

let $y = \cosh^{-1}(\sinh 2x)$

differentiating both sides with respect to x .

$$\frac{dy}{dx} = \frac{d \cosh^{-1}(\sinh 2x)}{dx}$$

$$= \frac{d \cosh^{-1}(\sinh 2x)}{d(\sinh 2x)} \times \frac{d \sinh 2x}{dx} \times \frac{dx}{dx}$$

$$= \frac{1}{\sqrt{\sinh^2 2x - 1}} \times \cosh 2x \times 2$$

$$= \frac{2 \cosh 2x}{\sqrt{\sinh^2 2x - 1}}$$

e) $e^{\cosh x/a}$

let $y = e^{\cosh x/a}$

differentiating both sides with respect to x .

$$\frac{dy}{dx} = \frac{d e^{\cosh x/a}}{dx}$$

$$= \frac{d e^{\cosh x/a}}{d \cosh x/a} \times \frac{d \cosh x/a}{dx} \times \frac{dx}{dx}$$

$$= e^{\cosh x/a} \times \sinh x/a \times \frac{1}{a}$$

$$= \frac{1}{a} e^{\cosh x/a} \times \sinh \frac{x}{a}$$

f) $e^{\tanh^{-1} x/a}$

let $y = e^{\tanh^{-1} x/a}$

differentiating on both sides with respect to x .

$$\frac{dy}{dx} = \frac{d e^{\tanh^{-1} x/a}}{dx}$$

$$= \frac{d e^{\tanh^{-1} x/a}}{d \tanh^{-1} x/a} \times \frac{d \tanh^{-1} x/a}{dx} \times \frac{dx}{dx}$$

$$= e^{\tanh^{-1} n/a} \times \frac{1}{1 - \frac{n^2}{a^2}} \times \frac{1}{a}$$

$$= \frac{1}{a} \times \frac{a^2}{a^2 - n^2} \times e^{\tanh^{-1} n/a}$$

$$= \frac{a}{a^2 - n^2} \times e^{\tanh^{-1} n/a}$$

∴

4) Find the derivatives of followings:

a) ~~$n^{\cosh n}$~~

let ~~$y = n^{\cosh n}$~~

differentiating on both sides with respect to ~~n~~ .

$$\frac{dy}{dn} = \frac{d}{dn} n^{\cosh n}$$

$$= \frac{d}{dn} n^{\cosh n} \times \frac{d \cosh n}{dn}$$

a) $n^{\cosh n}$

let $y = n^{\cosh n}$

Taking $\ln$ on both sides,

$$\ln y = \ln n^{\cosh n}$$

$$\Rightarrow \ln y = \cosh n \ln n$$

differentiating on both sides with respect to n ,

$$\frac{d \ln y}{dn} = \frac{d(\cosh n \cdot \ln n)}{dn}$$

$$\Rightarrow \frac{1}{y} \times \frac{dy}{dn} = \cosh n \frac{d \ln n}{dn} + \ln n \frac{d \cosh n}{dn}$$

$$\Rightarrow \frac{1}{y} \times \frac{dy}{dn} = \cosh n \times \frac{1}{n} + \ln n \times \sinh n$$

$$\Rightarrow \frac{dy}{dn} = \frac{y}{n} \cosh n + y \ln n \sinh n$$

$$\Rightarrow \frac{dy}{dn} = y \left(\frac{\cosh n}{n} + \ln n \sinh n \right)$$

$$\Rightarrow \frac{dy}{dn} = n^{\cosh n} \left(\frac{\cosh n + n \ln n \sinh n}{n} \right)$$

b) $n^{\sinh \frac{n^2}{a}}$

let $y = n^{\sinh \frac{n^2}{a}}$

Taking $\ln$ on both sides,

$$\ln y = \ln n^{\sinh \frac{n^2}{a}}$$

$$\Rightarrow \ln y = \frac{\sinh \frac{n^2}{a}}{a} \ln n$$

differentiating on both sides with respect to n .

$$\frac{d \ln y}{dn} = d \left(\frac{\sinh \frac{n^2}{a}}{a} \cdot \ln n \right)$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dn} = \frac{\sinh \frac{n^2}{a}}{a} \frac{d \ln n}{dn} + \ln n \frac{d \sinh \frac{n^2}{a}}{dn}$$

$$\Rightarrow \frac{1}{y} \times \frac{dy}{dn} = \frac{\sinh \frac{n^2}{a}}{a} \times \frac{1}{n} + \ln n \times \cosh \frac{n^2}{a} \times \frac{1}{a} \times 2n$$

$$\Rightarrow \frac{dy}{dn} = y \left(\frac{\sinh \frac{n^2}{a}}{n} + \frac{2n \ln n \cosh \frac{n^2}{a}}{a} \right)$$

$$\Rightarrow \frac{dy}{dn} = n^{\sinh \frac{n^2}{a}} \left(\frac{\sinh \frac{n^2}{a}}{n} + \frac{2n \ln n \cosh \frac{n^2}{a}}{a} \right)$$

c) $n^{\sinh^{-1} n}$

let $y = n^{\sinh^{-1} n}$

Taking $\ln$ on both sides,

$$\ln y = \ln n^{\sinh^{-1} n}$$

$$\ln y = \sinh^{-1} n \cdot \ln n$$

differentiating on both sides with respect to n .

$$\frac{d \ln y}{dn} = \frac{d(\sinh^{-1} n \cdot \ln n)}{dn}$$

$$\Rightarrow \frac{1}{y} \times \frac{dy}{dn} = \sinh^{-1} n \cdot \frac{d \ln n}{dn} + \ln n \cdot \frac{d \sinh^{-1} n}{dn}$$

$$\Rightarrow \frac{1}{y} \times \frac{dy}{dn} = \sinh^{-1} n \times \frac{1}{n} + \ln n \times \frac{1}{\sqrt{1+n^2}}$$

$$\Rightarrow \frac{dy}{dn} = y \left(\frac{\sinh^{-1} n}{n} + \frac{\ln n}{\sqrt{1+n^2}} \right)$$

$$\Rightarrow \frac{dy}{dn} = n^{\sinh^{-1} n} \left(\frac{\sinh^{-1} n}{n} + \frac{\ln n}{\sqrt{1+n^2}} \right)$$

d) $n^{\cosh^{-1} n/a}$

let $y = n^{\cosh^{-1} n/a}$

Taking $\ln$ on both sides,

$$\ln y = \ln n^{\cosh^{-1} n/a}$$

$$\Rightarrow \ln y = \frac{\cosh^{-1} n}{a} \ln n$$

differentiating on both sides with respect to n .

$$\frac{d \ln y}{dn} = \frac{d \left(\frac{\cosh^{-1} n}{a} \cdot \ln n \right)}{dn}$$

$$\Rightarrow \frac{1}{y} \times \frac{dy}{dn} = \frac{\cosh^{-1} n}{a} \cdot \frac{d \ln n}{dn} + \ln n \cdot \frac{d \cosh^{-1} n/a}{dn}$$

$$\Rightarrow \frac{dy}{dn} = y \left(\frac{\cosh^{-1} n/a}{n} \times \frac{1}{n} + \ln n \times \frac{1}{\sqrt{(n/a)^2 - 1}} \times \frac{1}{a} \right)$$

$$\Rightarrow \frac{dy}{dn} = n^{\cosh^{-1} n/a} \left(\frac{\cosh^{-1} n/a}{n} + \frac{\ln n}{a \sqrt{\frac{n^2 - a^2}{a^2}}} \right)$$

$$\Rightarrow \frac{dy}{dn} = n^{\cosh^{-1} n/a} \left(\frac{\cosh^{-1} n/a}{n} + \frac{\ln n}{\sqrt{n^2 - a^2}} \right)$$

e) $(\ln x)^{\sinh x}$

↳ let $y = (\ln x)^{\sinh x}$

⇒ ~~$y = \sinh x$~~

Taking ln on both sides,

$\ln y = \ln (\ln x)^{\sinh x}$

⇒ $\ln y = \sinh x \ln (\ln x)$

differentiating on both sides w.r.t x .

$\frac{d \ln y}{dx} = \frac{d (\sinh x \cdot \ln (\ln x))}{dx}$

⇒ $\frac{1}{y} \times \frac{dy}{dx} = \sinh x \frac{d \ln (\ln x)}{dx} + \ln (\ln x) \times \frac{d \sinh x}{dx}$

⇒ $\frac{1}{y} \times \frac{dy}{dx} = \sinh x \times \frac{1}{\ln x} \times \frac{1}{x} + \ln (\ln x) \times \cosh x$

⇒ $\frac{dy}{dx} = y \left[\frac{\sinh x}{x \ln x} + \cosh x \ln (\ln x) \right]$

⇒ $\frac{dy}{dx} = [\ln x]^{\sinh x} \left[\frac{\sinh x}{x \ln x} + \cosh x \ln (\ln x) \right]$

f) $(\sinh x)^{\cosh^{-1} x}$

↳ let $y = (\sinh x)^{\cosh^{-1} x}$

Taking ln on both sides,

$\ln y = \ln (\sinh x)^{\cosh^{-1} x}$

⇒ $\ln y = \cosh^{-1} x \ln (\sinh x)$

differentiating on both sides w.r.t x .

$\frac{d \ln y}{dx} = \frac{d (\cosh^{-1} x \cdot \ln (\sinh x))}{dx}$

⇒ $\frac{1}{y} \times \frac{dy}{dx} = \ln \sinh x \frac{d (\cosh^{-1} x)}{dx} + \cosh^{-1} x \frac{d \ln \sinh x}{dx}$

⇒ $\frac{dy}{dx} = y \left[\ln \sinh x \times \frac{1}{\sqrt{x^2 - 1}} + \cosh^{-1} x \times \frac{1}{\sinh x} \times \cosh x \right]$

⇒ $\frac{dy}{dx} = (\sinh x)^{\cosh^{-1} x} \left[\frac{\ln \sinh x}{\sqrt{x^2 - 1}} + \cosh^{-1} x \coth x \right]$

g) $(\cosh x)^{\cosh x}$

→ let $y = (\cosh x)^{\cosh x}$

Taking $\ln$ on both sides,

$$\ln y = \ln (\cosh x)^{\cosh x}$$

⇒ $\ln y = \cosh x \cdot \ln \cosh x$

differentiating on both sides w.r.t. x .

$$\frac{d \ln y}{dx} = \frac{d (\cosh x \cdot \ln \cosh x)}{dx}$$

⇒ $\frac{1}{y} \times \frac{dy}{dx} = \cosh x \frac{d (\ln \cosh x)}{dx} + \ln \cosh x \frac{d \cosh x}{dx}$

⇒ $\frac{dy}{dx} = y \left[\cosh x \times \frac{1}{\cosh x} \times \sinh x + \ln \cosh x \times \sinh x \right]$

⇒ $\frac{dy}{dx} = (\cosh x)^{\cosh x} [\sinh x + \sinh x \ln (\cosh x)]$

b) $\left(\tanh \frac{x}{a} \right)^{\ln x}$

→ let $y = \left(\tanh \frac{x}{a} \right)^{\ln x}$

Taking $\ln$ on both sides,

$$\ln y = \ln \left(\tanh \frac{x}{a} \right)^{\ln x}$$

⇒ $\ln y = \ln x \cdot \ln \left(\tanh \frac{x}{a} \right)$

differentiating on both sides w.r.t. x .

$$\frac{d \ln y}{dx} = \frac{d \left\{ \ln x \cdot \ln \left(\tanh \frac{x}{a} \right) \right\}}{dx}$$

⇒ $\frac{1}{y} \times \frac{dy}{dx} = \ln x \frac{d \ln \left(\tanh \frac{x}{a} \right)}{dx} + \ln \left(\tanh \frac{x}{a} \right) \frac{d \ln x}{dx}$

⇒ $\frac{dy}{dx} = y \left[\ln x \times \frac{1}{\tanh \frac{x}{a}} \times \operatorname{sech}^2 \frac{x}{a} \times \frac{1}{a} + \ln \left(\tanh \frac{x}{a} \right) \right]$

$$\Rightarrow \frac{dy}{dx} = \left(\tanh \frac{x}{a} \right)^{\ln x} \left[\frac{\ln x \times \cosh \frac{x}{a} \times 1}{a \sinh \frac{x}{a} \cosh^x \frac{x}{a}} + \frac{1}{x} \ln \left(\tanh \frac{x}{a} \right) \right]$$

$$\Rightarrow \frac{dy}{dx} = \left(\tanh \frac{x}{a} \right)^{\ln x} \left[\frac{2 \ln x}{a \sinh \frac{2x}{a}} + \frac{1}{x} \ln \left(\tanh \frac{x}{a} \right) \right]$$

$$\Rightarrow \frac{dy}{dx} = \left(\tanh \frac{x}{a} \right)^{\ln x} \left[\frac{2 \ln x}{a} \times \operatorname{cosech} \frac{2x}{a} + \frac{1}{x} \ln \left(\tanh \frac{x}{a} \right) \right]$$

i) $(\sinh^{-1} x + \cosh^{-1} x)^x$

Let $y = (\sinh^{-1} x + \cosh^{-1} x)^x$

Taking $\ln$ on both sides,

$$\ln y = \ln (\sinh^{-1} x + \cosh^{-1} x)^x$$

$$\Rightarrow \ln y = x \ln (\sinh^{-1} x + \cosh^{-1} x)$$

differentiating on both sides w.r.t x .

$$\Rightarrow \frac{d \ln y}{dx} = x \frac{d \ln (\sinh^{-1} x + \cosh^{-1} x)}{dx} + \ln (\sinh^{-1} x + \cosh^{-1} x) \times \frac{dx}{dx}$$

$$\Rightarrow \frac{1}{y} \times \frac{dy}{dx} = x \times \frac{1}{\sinh^{-1} x + \cosh^{-1} x} \times \left(\frac{1}{\sqrt{1+x^2}} + \frac{1}{\sqrt{x^2-1}} \right) + \ln (\sinh^{-1} x + \cosh^{-1} x) \times 1$$

$$\Rightarrow \frac{dy}{dx} = (\sinh^{-1} x + \cosh^{-1} x)^x \left[\frac{x}{(\sinh^{-1} x + \cosh^{-1} x)} \times \left(\frac{1}{\sqrt{1+x^2}} + \frac{1}{\sqrt{x^2-1}} \right) + \ln (\sinh^{-1} x + \cosh^{-1} x) \right]$$

⑤ Find $\frac{dy}{dx}$ if:

a) $x^{\sinh y} = y^{\cosh x}$

Taking $\ln$ on both sides,

$$\ln x^{\sinh y} = \ln y^{\cosh x}$$

$$\Rightarrow \sinh y \cdot \ln x = \cosh x \cdot \ln y$$

$\Rightarrow$ differentiating on both sides w.r. to x .

$$\frac{d(\sinh y \cdot \ln x)}{dx} = \frac{d(\cosh x \cdot \ln y)}{dx}$$

$$\Rightarrow \sinh y \frac{d \ln x}{dx} + \ln x \frac{d \sinh y}{dx} = \cosh x \frac{d \ln y}{dx} + \ln y \frac{d \cosh x}{dx}$$

$$\Rightarrow \sinh y \times \frac{1}{x} + \ln x \times \cosh y \frac{dy}{dx} = \cosh x \times \frac{1}{y} \times \frac{dy}{dx} + \ln y \times \sinh x$$

$$\Rightarrow \frac{\sinh y}{x} + \cosh y \cdot \ln x = \frac{\cosh x}{y} \times \frac{dy}{dx} + \ln y \cdot \sinh x$$

$$\Rightarrow \frac{\sinh y}{x} + \cosh y \cdot \ln x = \frac{\cosh x}{y} \times \frac{dy}{dx} + \ln y \cdot \sinh x$$

$$\Rightarrow \sinh y \times \frac{1}{x} + \cosh y \cdot \ln x \times \frac{dy}{dx} = \cosh x \times \frac{1}{y} \times \frac{dy}{dx} + \ln y \times \sinh x$$

$$\Rightarrow \cosh y \cdot \ln x \times \frac{dy}{dx} - \cosh x \times \frac{1}{y} \times \frac{dy}{dx} = \ln y \times \sinh x - \frac{\sinh y}{x}$$

$$\Rightarrow \frac{dy}{dx} (\cosh y \cdot \ln x - \cosh x \times \frac{1}{y}) = (\ln y \times \sinh x - \frac{\sinh y}{x})$$

$$\Rightarrow \frac{dy}{dx} \left(\frac{y \cosh y \cdot \ln x - \cosh x}{y} \right) = \frac{1}{x} (\ln y \cdot \sinh x - \sinh y)$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} \frac{(\ln y \cdot \sinh x - \sinh y)}{(y \cosh y \cdot \ln x - \cosh x)}$$

Orient

b) $x^2 + y^2 = \tanh(xy)$

→ ~~Integ~~ Differentiating on both sides with respect to x .

$$\frac{dx^2}{dx} + \frac{dy^2}{dx} = \frac{d \tanh(xy)}{dx}$$

$$\Rightarrow 2x + 2y \times \frac{dy}{dx} = \text{sech}^2 xy \times \left[x \cdot \frac{dy}{dx} + y \times \frac{dx}{dx} \right]$$

$$\Rightarrow 2x + 2y \times \frac{dy}{dx} = \text{sech}^2 xy \times \left[x \times \frac{dy}{dx} + y \right]$$

$$\Rightarrow 2x + 2y \times \frac{dy}{dx} = x \text{sech}^2 xy \times \frac{dy}{dx} + y \text{sech}^2 xy$$

$$\Rightarrow 2y \times \frac{dy}{dx} - x \text{sech}^2 xy \times \frac{dy}{dx} = y \text{sech}^2 xy - 2x$$

$$\Rightarrow \frac{dy}{dx} (2y - x \text{sech}^2 xy) = y \text{sech}^2 xy - 2x$$

$$\Rightarrow \frac{dy}{dx} = \frac{y \text{sech}^2 xy - 2x}{2y - x \text{sech}^2 xy}$$

c) $xy = \coth^{-1}(x+y)$

→ differentiating on both sides w.r.t x .

$$\frac{d(xy)}{dx} = \frac{d(\coth^{-1}(x+y))}{dx}$$

$$\Rightarrow x \times \frac{dy}{dx} + y \times \frac{dx}{dx} = \frac{-1}{(x+y)^2 - 1} \times \left(\frac{dx}{dx} + \frac{dy}{dx} \right)$$

$$\Rightarrow x \times \frac{dy}{dx} + y \times 1 = \frac{-1}{(x+y)^2 - 1} \times \left(1 + \frac{dy}{dx} \right)$$

$$\Rightarrow x \times \frac{dy}{dx} + y = \frac{-1}{(x+y)^2 - 1} - \frac{1}{(x+y)^2 - 1} \times \frac{dy}{dx}$$

$$\Rightarrow \left(x + \frac{1}{(x+y)^2 - 1} \right) \times \frac{dy}{dx} = \frac{-1}{(x+y)^2 - 1} - y$$

$$\Rightarrow \frac{(x(x+y)^2 - x + 1) \times \frac{dy}{dx}}{(x+y)^2 - 1} = \frac{-1 - y(x+y)^2 + y}{(x+y)^2 - 1}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1 - y + y(x+y)^2}{1 - x - x(x+y)^2 - 1}$$

$$\therefore \frac{dy}{dx} = \frac{1 - y + y(x+y)^2}{x - x(x+y)^2 - 1}$$